

NEHARIKA YARASU

Chirala, AP | +91 9502511485 | yarasuneharika@gmail.com | LinkedIn: <https://www.linkedin.com/in/yarasu-niharika-11108126>

GitHub: <https://github.com/harika2028>

PROFESSIONAL SUMMARY

Final-year ECE undergraduate (2026) specializing in VLSI Design, RTL Development, UVM basics, TestBench Development, Functional Verification, and Low-Power Digital Circuits with proven expertise in Verilog HDL, Xilinx Vivado, DSCH, and Microwind for ASIC/FPGA design flows. Skilled in RTL Coding, Synthesis, Timing Analysis, CDA/STA Debugging and Digital Logic Design with a strong focus on power-efficient architectures and tapeout-ready simulations. Patent filer (IPO 2026) and ICCET 2026 paper presenter, showcasing research aptitude, innovation, and technical documentation skills sought by semiconductor recruiters. Actively seeking VLSI / RTL Design Engineer roles to drive front-end design, verification, & low-power SoC development in the semiconductor domain.

EDUCATION

B.Tech, Electronics & Communication Engineering

2022 – 2026

Bapatla Women's Engineering College, Andhra Pradesh | CGPA: 8.5 / 10 | Coursework: Digital Logic Design, CMOS Technology, VLSI, DSP, Computer N/Ws

INTERNSHIP EXPERIENCE

Manufacturing & Processing Intern | Tata Electronics Pvt. Ltd.

2026 | On-site

- Observed semiconductor manufacturing workflows and quality control processes; developed understanding of **ASIC design flow** and production-level testing standards

Virtual VLSI Design Intern | CodTech IT Solutions

2025 | Remote

- Performed **RTL design and digital circuit simulation** using Xilinx Vivado; implemented low-power VLSI optimization techniques reducing design power by ~15%
- Developed **functional verification testbenches** and conducted waveform analysis to validate digital circuit behavior across multiple design iterations
- Simulated combinational and sequential circuits including **FSMs, MUX/Encoders/Decoders, and Arithmetic Circuits** using EDA Playground and ModelSim

IoT & Embedded Systems Intern | SRC e-Solutions

2024 | On-site

- Designed **PCB layouts** and worked on drone technology using ESP32 and Flight Controller; contributed to UAV system integration and testing
- Interfaced sensors and microcontrollers for embedded control systems; gained hands-on experience in **hardware-software co-design**

PROJECTS

Advanced 32-Bit Hybrid Parallel Prefix Adder | Verilog HDL | Xilinx Vivado

Jan – Mar 2026

- Designed Hybrid Han-Carlson + Ladner-Fischer PPA achieving **~15% area, ~18% power, and ~12% delay reduction** over conventional PPA via RTL optimization and timing analysis

Low-Power PPA using GDI & Transmission Gate | DSCH | Microwind

Oct – Dec 2025

- Implemented transistor-level design using **GDI, MGDI, FSGDI, and Transmission Gate** techniques; reduced power, area, and transistor count vs. CMOS; evaluated Power-Delay Product (PDP) via DSCH/Microwind

Digital Locker System | 4:1 MUX | Verilog HDL | DSCH | Vivado

Aug – Sep 2025

- Built password-based access control using **4:1 MUX architecture**; created gate-level schematics in DSCH; verified design via RTL simulation and waveform analysis

Touch Me Not Crop Protection System | IoT & Embedded Systems Arduino Uno

May – Aug 2025

- IoT-based smart farming system using ESP8266, IR sensors, and soil moisture sensors for crop monitoring and intrusion detection. Solar tracking increased energy utilization by ~30% for efficient farm management.

Speech Signal Denoising System | DSP | MATLAB

Mar – Sep 2024

- Implemented DSP-based speech enhancement in MATLAB using filtering and noise reduction techniques, improving SNR by 20–25%.

Ongoing: CPU Design | Traffic Light Controller (FSM) | AI Accelerator Architecture | Quantum VLSI (IBM Qiskit)

TECHNICAL SKILLS

VLSI & Design: RTL Design, ASIC Design Flow, FPGA Design, Digital Logic Design, CMOS Technology, Low-Power VLSI, FSM Design, Timing Analysis

HDL / Languages: Verilog HDL, VHDL, SystemVerilog (Basics), Python, C, MATLAB, Linux Commands (learning)

Tools: Xilinx Vivado, DSCH, Microwind, ModelSim, EDA Playground, Logisim, IBM Qiskit & QASM, Linux

Verification: Functional Verification, Testbench Development, Simulation & Debugging, Waveform Analysis

Concepts: MUX, Encoders, Decoders, Arithmetic Circuits, Parallel Prefix Adders, GDI, Modified GDI, Full Swing GDI, Quantum Computing Basics

PUBLICATION & PATENT

- Patent (IPO, Apr 2026):** Design of a Low Power Hybrid Parallel Prefix Adder for VLSI Applications | App No. 202641040687
- ICCET 2026 Paper Presenter:** Low-Power VLSI Parallel Prefix Adder Architectures - peer-reviewed international conference

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL:** CMOS VLSI Fundamentals | Optical Wireless Communication Beyond 5G | Cryogenic Electronics for Quantum Computing
- Coursera:** Nanotechnology & VLSI | **Achievements:** 1st Prize - Project Expo 2025 | MSME Hackathon 2.0 (IoT Agriculture) | NCC Cadet | IETE Member